

Evidentiary reasoning within the Analytical Hierarchy Process: Use of graph entropy to rank decision elements

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Abstract

Decision makers often require insight into the relative strength of the elements of evidence on which their choices hinge. The Analytic Hierarchy Process (AHP) contains innate mechanisms leading to the rigorous quantification of elements' relative strength. Comparative judgements ("pairwise comparisons") produce lists of edges for a complete graph. Under certain conditions, such graphs are also directed and acyclic, belonging to the class of tournament graphs. Intensities of Importance from comparative judgements provide weights for graphs' edges. Relative-entropy calculations based on Laplacians for induced subgraphs quantify the relative evidentiary strength associated with the decisional evidence omitted from them. A previously published case study related to an offshore-pipeline material selection demonstrates the approach. The graph-entropy method provides greater strength-of-contribution contrast than sensitivity analysis from the case study.

Introduction.

Decision-making nearly always occurs under conditions of incomplete or imperfect information. No less than Nobel-prize economist Frederick von Hayek recognized this:

"...I prefer true but imperfect knowledge, even if it leaves much undetermined and unpredictable, to a pretence of exact knowledge that is likely to be false..." (emphasis added) (von Hayek (1974), cited in Kay & King (2020)).

The treatment of uncertainty in quantitative decision making represents a formidable research challenge confronting multiple facets. We conventionally decompose uncertainty into epistemic and aleatory (or stochastic) components (Salicone & Prioli (2018)). Sound evidentiary reasoning must account for the consequences of both.

The present study focuses primarily on the epistemic. Hamlett (2019, 2025) argues for the primacy of resolving epistemic uncertainty in business decision making. Exemplified by Zellweger & Zenger (2023); Rindova

& Martins (2024); Townsend, et al (2025); Ehrig, et al (2026), this perspective holds sway in the entrepreneurial and strategic decision-making communities. Jackson (2019) corroborates from a philosophy-of-science point of view.

Motivation: How strong is our evidence?

Ehrig, et al (2026) argue that entrepreneurial decision making in particular inescapably requires "...logic and reasoning with the aid of very small samples of data" (emphasis added). Circumstances demanding this sort of reasoning arise in numerous contexts beyond new-business formation (Martin & Olsberg (2007); Eisenmann (2013); Swanson (2024)). Fundamentally, entrepreneurial ventures amount to epistemic explorations (Camuffo, et al (2024)). Howard & Abbas (2016) devoted considerable attention to the adverse effects of imperfect information — an endemic condition within entrepreneurship — when trying to select optimum scenarios.

Nontrivial challenges inevitably arise even when con-

siderable amounts of statistically coherent, phenomenologically illuminating data exist. Cover & Thomas (2006) derive an information-theoretic data-processing inequality: “...no processing of Y ... can increase the information that Y contains about X .” Corroborating from a statistical-learning perspective, Hastie, Tibshirani, Friedman (2009) define *irreducible variance*. Kay & King (2020) warn of pervasive nonstationarity in real-world problems: Statistical-learning training data based on past experience might not represent future conditions very well. More generally, Townsend, et al (2025) articulate the case for “...computationally irreducible barriers to the predictive capabilities and task performance advantages of AI systems...” for many tasks.

Extending Dempster’s (1967) seminal concept, Shafer (1976) offered a rigorous approach to quantifying epistemic uncertainty in evidentiary reasoning. More-recently, Shenoy (2023) applied Shafer’s reasoning to quantify evidentiary incompleteness in Bayesian networks, a causal-inference methodology commonly employed within the entrepreneurial decision-making community (Camuffo, et al (2024); Ehrig, et al (2026)).

Hué, et al (2026) use a stochastic-detection area-under-the-curve (AuC) approach (cf. Fawcett (2026)) to ascertain evidentiary elements’ relative strengths. Resembling stepwise regression (Olive (2017)), their approach also exhibits conceptual parallels to Shafer (1976). Cox (2026a,b) offers an emerging approach to uncertainty characterization based on sheaf theory (Rosiak (2022)).

Why the Analytical Hierarchy Process (AHP)?

AHP provides an axiomatic framework (Saaty (1986)) for “...logic and reasoning with the aid of very small samples of data” (Ehrig, et al (2026)). It is a multi-criterion decision methodology (MCDM) enjoying widespread applicability to group decision making (Saaty (1994)), risk analysis (e.g., Saaty (1987); Vaidya & Sushil (2006); Linkov, et al (2006)), and other problems. AHP is often applied to situations where substantial amounts of statistically coherent measurements are not immediately available to support more-empirical approaches.

The AHP methodology begins with identifying a plausible evidentiary span: The range of measurable factors — elements in the vernacular of Saaty (1986, 1994) — believed to differentiate within the range of scenarios under consideration. A *hierarchy model* represents this as a tree graph. The responsible individual decision maker or a panel of experts then performs comparative judgements: “...the numerical representation of a relationship between two elements that share a common parent” in the tree graph (Saaty (1994)). A case study

subsequently demonstrates AHP’s use to rank plausible alternatives.

Not strictly a stand-alone method, AHP can be fused with other approaches. Bayesian techniques allow for subsequent revision of the comparative judgements as stronger empirical evidence emerges (Merrick, van Dorp, Singh (2005); Li, H., et al (2025)). Li, J. et al (2025) combined back-propagation neural networks with the AHP to design shields for neutron generators. Xiao, et al (2025) explored simulation-optimization approaches to circumvent altogether the elicitation of comparative judgements from experts. Lukinskiy, Lukinskiy, Bazhina (2025); Sáenz-Royo & Chiclana (2026) studied methods to improve the robustness of expert inputs into comparative judgements. Syed, Shabarchin, Lawryshyn (2020) describe AHP’s use in eliciting prior probabilities for Bayesian reliability analysis.

Evidentiary-strength analysis using the AHP.

That comparative judgements for an AHP hierarchy model define a complete graph provides the seminal insight underlying this investigation. Additional opportunities arise when the resulting graph is also directed and acyclic. Saaty (1986) recognized this: “Partially ordered sets with a finite number of elements can be conveniently represented by a directed graph”. Complete DAGs belong to a special class of graphs called *tournament graphs* (Moon (2015)). Nishizawa (1999) obliquely considered graphical methods for AHP consistency. The decision-theory literature however contains scant exploration of AHP’s graphical characteristics and opportunities arising therefrom.

Entropy — a key concept from information theory — provides an abstract but rigorous framework for characterizing evidentiary strength (Cover & Thomas (2006)). In general, “the entropy of a graph is interpreted as its *structural information content* and serves as a complexity measure” (emphasis added) (Dehmer & Mowshowitz (2011); Mowshowitz & Dehmer (2012)). Structural information content bears direct relevance to the present concern of evidentiary reasoning. Brown, et al (2020) studied the particular characteristics of tournament graphs.

What questions do entropies of AHP-based graphs allow us to answer?

Endogenous to the AHP formulation itself, *relative entropy* quantifies the marginal informational contribution by a subset of elements within the evidentiary span. This addresses the objective pursued by Hué, et al (2026). Graph-entropy measures however do not require extensive empirical data spanning of both positive and negative cases, as with Hué’s approach. Congruent with the AHP methodology explored

here, Jiroušek, Kratochvíl, Shenoy (2023) use entropy of belief-function graphical models to decompose their elements' relative evidentiary strength.

By what mechanism do we infer the differential information between hierarchically-adjacent AHP hierarchy models? Relative entropy provides a well-understood approach (Cover & Thomas (2006)). Relative entropy for graphs has been well-researched (Wong & You (1985); Liu, et al (2022); Guanlei, Xiaogang, Xiaotong (2025). Brown, et al (2020) explored entropy differences between same-order tournament graphs with differing directional structures. In our case, the relative entropy of the graph associated with the overall evidentiary span with respect to a constituent induced subgraph quantifies the information lost by omission of elements from the latter.

What innovation is required to define a methodology for AHP-based evidentiary-strength analysis? First, we must formulate an entropy definition for complete, mixed graphs with weighted edges. Brown, et al (2020) analyzed directed graphs with unweighted (uniformly unit-weighted) edges. Chen, et al (2015), Kuo (2022), and Liu, et al (2022) formulate entropies for regular (non-directional) graphs comprised of weighted edges. The intensities of importance from the AHP comparative judgements provide graph-edge weights used here.

Second, relative entropies of hierarchically adjacent hierarchy models must be coupled. The *Hierarchy* term in AHP refers to a hierarchic deconstruction of decisions into constituent elements. The AHP method produces a set of decision-element comparative judgements — represented by a tournament graph — for each hierarchy model. (Cover & Thomas, 2006, §2.5) describe chain rules for entropy, relative entropy, and mutual information. Such mechanisms offer a rigorous approach to link hierarchically adjacent tournament graph.

Why is “heavy” machinery of graph entropy called-for? We might intuitively expect that individual comparative judgements suffice to inform us about the most-influential elements. The AHP's machinery however accounts for “mutual coupling” between elements: “...many of the concepts derived for hierarchies also relate to systems with feedback” (Saaty (1986)). That comparative judgements define a graph — a network — provides the basis of its power. Our graph-entropy machinery accounts for the aggregate effect of all the evidence informing a decision or risk assessment. Individual importance intensities fail to provide this.

Decision Quality as a check on the completeness of evidentiary reasoning.

Decision Quality (DQ) represents a prominent applied decision-analysis framework that emphasizes evidentiary reasoning (Spetzler, Winter, Meyer (2016)). Philosophically congruent with design-of-experiments practice (Cox & Reid (2000)), DQ provides a de facto standard of practice for the Society of Decision Professionals (SDP). SDP is a community of applied decision analysts and scientists. Introducing DQ here — independent of the mechanics of AHP — reinforces our logical completeness.

The pattern of reasoning underlying Saaty's (1986) AHP methodology loosely maps to the DQ framework. Figure 1 depicts the DQ framework, comprised of six links.

- Establishing an *Appropriate Frame* specifies the scope of the decision: What specifically is to be decided. Thoughtful framing combines prescriptive with proscriptive judgements to provide analytical clarity. Saaty (1994) describes this as “...the most creative part of decision making that has a significant effect on the outcome...”.
- By definition, the act of deciding involves selection of one or more options from a discrete, precisely defined set of *Creative Alternatives*.
- *Relevant and Reliable Information* constitute the evidence weighed in selecting the best alternative.
- *Clear Values and Tradeoffs* specify the criteria by which the best alternative is identified. They specify the decision makers' values, clarifying why the best alternative actually is the best.
- *Sound Reasoning* defines the logic employed in identifying the best alternative given the available evidence. Creighton (1909) cautioned that “...it is not possible to lay down definite rules for the guidance of thinking in every case.” The reasoning approach must therefore fit the circumstances resulting from the links above. AHP is itself a distinct reasoning approach.
- *Commitment to Action* represents the follow-through in implementing or adopting the selected alternative. Absent a commitment to adopt or act upon the best alternative, a decision arguably has not occurred. The analysis was academic. Persistence with the status quo nearly always occurs among the alternatives under consideration. Such a selection should however result from applying sound reasoning to the relevant available evidence.

Figure 1: Decision-Quality framework. (From Society of Decision Professionals (DQ))



Now, use of AHP by no means suffices to achieve DQ. Thoughtful application of AHP does address a subset of DQ's links. Injudiciously applied, it does not however guarantee satisfaction of the von Neumann-Morgenstern axioms (Kochenderfer, 2015, p. 58) undergirding decision science. Featuring AHP as an approach to DQ in no way constitutes our objective here. We instead explore AHP's ability to highlight how subsets of decisional values and spans of evidence influence the results of sound reasoning.

Course of investigation.

We begin with a brief illustration of AHP methodology. A real-world case study concisely demonstrates its application to preference-ranking a set of scenarios. The actual ranking lies beyond our specific interest, but is included for completeness. The complete graphs resulting from the comparative judgements provide our point of departure into a novel methodology.

We then calculate the entropies for our comparative-judgement graphs associated with distinct hierarchy models. This extends Brown, et al (2020) through adoption of a Laplacian structure that includes edge weights and unit-weight undirected edges. Our AHP-graph Laplacian fuses the structure of a conventional unit-edge-weighted directed-graph Laplacian with comparative-judgement importance intensities, which provide the edge weights. This incorporates infor-

mation about degree of influence in the interconnected evidentiary framework.

Finally, we calculate entropies and relative entropies from our AHP-based graph structure comprised of multiple, hierarchically coupled hierarchy models. We interpret the results for our real-world case study. Approaches analogous to those by Jiroušek, Kratochvíl, Shenoy (2023) guide our thinking. We also explore the relationship between importance intensities and relative entropy. This quantifies relative strengths of contribution by subsets of evidentiary elements.

AHP case study .

Zeng, D., et al (2025) employed AHP to select materials from which to construct pipelines for use in an offshore oil and gas project. Their analysis considers material-mechanical and -chemical technical details that are tangential to our purposes. The AHP mechanics primarily interest us.

Zeng, D., et al (2025) particularly suits the purposes of this investigation for three reasons.

1. It demonstrates a real-world application. Introductory AHP presentations often employ contrived cases. A real-world case study provides a more-informative and -convincing demonstration.
2. This case is recent. Given that the AHP literature

spans decades, a reader might succumb to the temptation to conclude that the method is anachronistic. Our case here demonstrates the application of AHP to a contemporary design problem for an economically essential application.

3. It strikes a balance between tractability and complexity. Many pedantic demonstrations are borderline-trivial, constructed to illustrate AHP mechanics. The present investigation benefits from a moderately greater degree of complexity. The selected case provides the opportunity to demonstrate the use of relative entropy to couple hierarchically adjacent hierarchy models. That it considers nine distinct criteria also makes it interesting.

The following offers neither criticism nor endorsement of Zeng, D., et al (2025). Their work merely serves to illustrate AHP mechanics.

We strive in the following to select AHP terminology that remains close to Saaty (1986, 1994). Variations in terminology occur throughout the AHP-application literature. For example, the quantity Saaty (1986) defines as a *priority vector* is frequently referred to as a “weight vector” or “weights”. We employ *priority vector* here.

Specification of the evidentiary span.

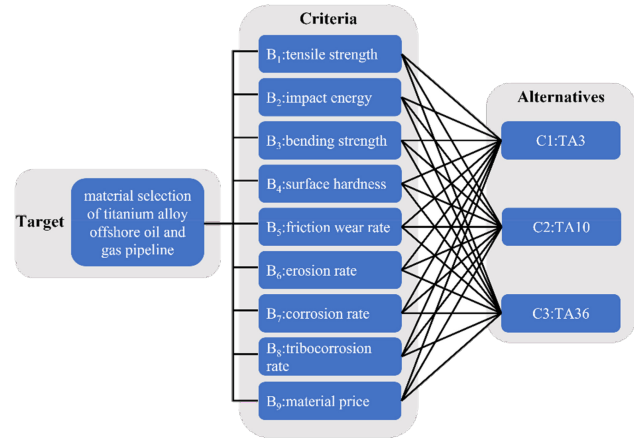
Figure 2 reproduces the *hierarchy model* employed by Zeng, D., et al (2025) for their pipeline-material selection. It depicts a hierarchically structured decision. The identified target describes a decisional activity: “Material selection of titanium alloy ...”. Their nine criteria include eight mechanics-of-material characteristics, $B_1, \dots, B_8$, and one economic one B_9 . The latter suggests an economic basis for utility.

The constituent criterion-to-alternative hierarchy models address how each of the candidate chemical compositions address the alternatives. Zeng, D., et al (2025) describe an extensive material-testing regime to ascertain the the performance of each with respect to the material-mechanical criteria. Although comparative judgments introduce subjective aspects to the material-selection decision, Zeng, D., et al (2025) also bring to bear extensive empirical experimental evidence.

Visual, tree-graph depiction of the decision-element hierarchy aids in understanding by human decision-makers. Idiosyncratic style notwithstanding, Figure 2 follows the ubiquitous pattern originating with Saaty (1986, 1994), and appearing throughout the MCDM literature. Structural elements of Figure 2 map to key DQ links from Figure 1.

- The rectangular box to the left “Target: material selection ...” at the left represents the Goal or Focus,

Figure 2: Hierarchy model used for offshore-pipeline material-selection case study. (From Zeng, D., et al (2025), Figure 9)



in Saaty’s (1986) terms. A clear goal or focus results from thoughtful *Framing*, in DQ terms.

- The horizontally centered box identifies *Criteria*. Corresponding to DQ’s Values, these define the span of measurable criteria believed to materially distinguish between alternatives. Zeng, D., et al (2025) identified nine such criteria.
- Figure 2 explicitly depicts the *Alternatives* under consideration. That only three — distinguished by labels TA_3 , TA_{10} , and TA_{36} , each corresponding to a distinct chemical composition (Zeng, D., et al (2025) Table 1) — are considered makes their explicit representation here practical. When the number of alternatives is large — such as analyzing the risk of a large portfolio of distinct assets — showing all of them clutters the hierarchy-model depiction.

The target-to-criterion hierarchy model in Figure 2, spanning nine elements, introduces a modicum of complexity. Saaty (1986) specifies a *transitive-consistency* criterion that, analogous to but stronger than the van Neumann-Morgenstern transitivity axiom (Kochenderfer, 2015, p. 58)

$$(A \succ B) \wedge (B \succ C) \Rightarrow A \succ C \quad (1)$$

proscribes circular reasoning.

Enforcing this constraint during the comparative-judgement process can prove challenging, particularly as the number of hierarchy-model elements grows. Conventionally, AHP practitioners calculate a *Consistency Index*, which algebraically quantifies the degree to which this transitivity is satisfied (Pant, et al (2022)). Representing comparative judgements as a directed

graph provides an unambiguous mechanism to identify violations of von Neumann-Morgenstern transitivity.

Alignment of AHP to mainstream decision science involves specification of a utility function. AHP goals do not explicitly represent utilities in the conventional, decision-science sense. AHP practitioners can impose such an association exogenous to the mechanics of the methodology itself. Zeng, D., et al (2025) speak to utility qualities, but do not explicitly articulate a utility function or theory.

Comparative judgements about relative importance of criteria.

AHP imbues values and tradeoffs into decision-making through comparative judgements: “...a matrix to carry out *pairwise comparisons* of the relative importance of the elements [criteria]...” (emphasis added) (Saaty (1986)). *Pairwise* is the operative term, here. Practically, one assembles a list of unique, unordered element pairs for each hierarchy model. Conventionally, explicit representation of hierarchy-element comparative judgements is skipped. The results applied directly to a reciprocal matrix. Since these results provide our excursion into graph-based evidentiary reasoning, we pay particular attention here.

Comparative-judgement principles.

If the hierarchy model for a given objective (or criterion) spans n elements, then $\binom{n}{2}$ element pairs must be considered. The cardinality $\binom{n}{2}$ specifies the range of hierarchy-model element pairs that must be considered.

Application of comparative judgement imposes an ordering that must subsequently be observed. Consider a hierarchy model $\mathfrak{H}$ comprised of n elements, $\mathfrak{H} \supseteq \{e_1, \dots, e_n\}$. Comparative judgement $\mathcal{J}_{\mathfrak{H}}$ entails $\binom{n}{2}$ distinct mappings

$$(e_i, e_j) : \mathcal{J}_{\mathfrak{H}} \mapsto a_{i,j} \quad \forall i \neq j, \quad (2a)$$

where $a_{i,j}$ denotes a corresponding element of a reciprocal matrix. The mapping $\mathcal{J}_{\mathfrak{H}}$ imposes

$$\begin{aligned} e_i \succ e_j &\Rightarrow a_{i,j} \geq 1 \\ e_i \prec e_j &\Rightarrow a_{i,j} < 1 \end{aligned} \quad (2b)$$

Consequently, after its application, element ordering must be maintained. Finally, $a_{i,j}$ is positive-definite.

Material-selection comparative judgements.

Figure 2 contains ten hierarchic models: One target-to-criterion model comprised of nine elements; and nine criterion-to-alternative models, each with three

elements. Consequently, $\binom{9}{2} = 36$ element pairs must be considered for the target-to-criterion model. Each criterion-to-alternative model entails $\binom{3}{2} = 3$ element pairs. Each element pair corresponds to an edge for the graph by which the hierarchy model is represented.

Table 1a contains the comparative judgements assigned by Zeng, D., et al (2025) for each element pair depicted in the target-to-criterion hierarchy model in Figure 2. Table 1b contains element-pair judgements for each of the criterion-to-alternative models. We defer construction of corresponding reciprocal matrices to the next subsection. Explicit depiction of the comparative judgements here contributes to the clarity of present course of reasoning. In particular, the edge lists in Tables 1a and 1b provide a key of departure for subsequent extension of the AHP methodology.

Assignment of intensities of importance.

Now, Saaty (1986) Table 1 and Saaty (1994) Table 1 offer a “fundamental scale” containing heuristics for assignment of *intensity-of-importance* values to hierarchy-model element pairs. Saaty (2001b) offers a psychophysics justification for this scale, and argues for an upper bound of 9 when assigning values to importance intensities. Zeng, D., et al (2025) employ this scale (their Table 7), as do many other AHP applications appearing in the literature. Saaty (1986), the most-rigorous description of the AHP methodology, contains no explicit prescription for this schema for importance-intensity assignment.

Saaty (1986, 2001b) do however speak to preferential ordering, transitive consistency, and homogeneity considerations. Consequently, alternative, more-rigorous importance-intensity assignments can be considered. Application of the personal-probability indifference principle arises as one option (Savage (1972); de Finetti (1974)). Hubbard & Seiersen (2023) — addressing the specific domain of cybersecurity — describe more-empirical approaches to measurement. Application of such alternatives in lieu of Saaty’s “fundamental scale” deflects a common critique of AHP’s logical rigor.

Construction of the comparative-judgement table.

The construction of Tables 1a and 1b anticipates their use in defining directed graphs in the sequel. The AHP literature — Saaty (1986, 1994, 2003), in particular — prescribes nothing about the ordering of element pairs in comparative judgement. Construction of the tables follows the following convention.

- Populate the “intensity of importance” column with the value greater than or equal to unity.

Table 1a: Comparative judgements for each element pair within the target-to-criterion hierarchy model of Figure 2. (After Zeng, D., et al (2025), Table 10.)

reference element	comparative element	intensity of importance
B_1	B_2	4
B_1	B_3	2
B_1	B_5	4
B_1	B_6	2
B_1	B_7	2
B_1	B_9	2
B_2	B_4	3
B_2	B_6	1
B_2	B_9	1
B_3	B_2	3
B_3	B_4	3
B_3	B_5	2
B_3	B_6	2
B_3	B_7	2
B_3	B_9	3
B_5	B_2	3
B_5	B_4	3
B_5	B_6	1
B_5	B_7	2
B_5	B_9	2
B_6	B_4	2
B_6	B_7	5
B_6	B_9	2
B_7	B_2	3
B_7	B_4	3
B_7	B_9	2
B_8	B_1	5
B_8	B_2	5
B_8	B_3	5
B_8	B_4	5
B_8	B_5	5
B_8	B_6	5
B_8	B_7	5
B_8	B_9	9
B_9	B_4	1

Table 1b: Comparative judgements for each element pair within each of the criterion-to-alternative hierarchy models in Figure 2. (After Zeng, D., et al (2025), Table 8).

hierarchy model	reference element	comparative element	intensity of importance
B_1	TA_3	TA_{10}	3
B_1	TA_{36}	TA_3	3
B_1	TA_{36}	TA_{10}	5
B_2	TA_{10}	TA_3	3
B_2	TA_{36}	TA_3	5
B_2	TA_{36}	TA_{10}	3
B_3	TA_3	TA_{10}	3
B_3	TA_{36}	TA_3	3
B_3	TA_{36}	TA_{10}	5
B_4	TA_3	TA_{10}	2
B_4	TA_{36}	TA_3	3
B_4	TA_{36}	TA_{10}	3
B_5	TA_3	TA_{10}	3
B_5	TA_{36}	TA_3	3
B_5	TA_{36}	TA_{10}	7
B_6	TA_3	TA_{10}	2
B_6	TA_{36}	TA_3	3
B_6	TA_{36}	TA_{10}	3
B_7	TA_3	TA_{10}	3
B_7	TA_3	TA_{36}	5
B_7	TA_{10}	TA_{36}	3
B_8	TA_3	TA_{10}	2
B_8	TA_{36}	TA_3	3
B_8	TA_{36}	TA_{10}	5
B_9	TA_3	TA_{10}	3
B_9	TA_{36}	TA_3	3
B_9	TA_{36}	TA_{10}	5

- Assign to the “reference element” column the most-preferred element, associated with the preferred intensity-of-importance value. This element subsequently corresponds to the head of the directed graph edge connecting the two elements.
- Populate the “comparative element” column the less-preferred element. This element corresponds to the tail of the directed-graph edge.

This convention imbues two conveniences into the procedure. First, the structure of the table endogenously contains the directive structure of the graph. This simplifies the algorithmic logic used to construct the resultant tournament graph. Second, it inclines us to think of importance intensities in terms as personal probabilities (Savage (1972); de Finetti (1974)) in lieu of the arguably problematic “fundamental scale”.

Comparative-judgement indifference and its consequences.

That Table 1a contains unit-valued importance intensities — hierarchy-model element pairs regarding which comparative-judgement experts were indifferent — raises both interest and a challenge. Four undirected edges appear. This introduces an aspect of complexity. Unit-valued importance intensities imply undirected edges. In such cases, *mixed graphs* occur: Graphs containing both directed and undirected edges.

We subsequently handle this in construction of the graph Laplacians, on which entropy calculations are based. Our Laplacians must account for both directed and undirected edges, as well as other-than-unity edge weights. Aluded to previously, graph-Laplacian constructions for distinct graph classes appear separately in the literature (Wong & You (1985); Dehmer & Mowshowitz (2011); Mowshowitz & Dehmer (2012); Brown, et al (2020); Liu, et al (2022); Guanlei, Xiaogang, Xiaotong (2025)). Judiciously combining graph-class-specific Laplacian structures with our mixed-graph structures represents the task of the present investigation.

Transformation of comparative judgments into priority vectors.

AHP priority vectors provide the key quantitative mechanism for evidentiary aggregation in MCDM applications. Mechanically, this is accomplished by constructing a *reciprocal matrix* from the comparative judgements for each hierarchy model. A reciprocal matrix is a Hermetian matrix with unit-valued diagonals and intensity-importance values assigned to off-diagonal elements.

Construction of reciprocal matrices from comparative judgements.

As the name implies, a reciprocal matrix is a matrix of comparative judgements within which upper-diagonal elements are reciprocals of the lower. Consider again a hierarchy model $\mathfrak{H}$ comprised of n elements, $\mathfrak{H} \supseteq \{e_1, \dots, e_n\}$. Then the corresponding reciprocal matrix $A_{\mathfrak{H}}$ becomes

$$A_{\mathfrak{H}} = \begin{bmatrix} 1 & a_{1,2} & \dots & a_{1,n} \\ a_{2,1} & 1 & \dots & a_{2,n} \\ \vdots & \ddots & 1 & \vdots \\ a_{n,1} & a_{n,2} & \dots & 1 \end{bmatrix} \quad (3a)$$

subject to

$$a_{i,j} = a_{j,i}^{-1}, \quad (3b)$$

and where (e_i, e_j) are assigned

Weight-vector calculation.

Principal-eigenvector method.

Geometric-mean method.

Entropy-weight method. For the O-C model in Table 1a, this appears as

$$A_{O-C} = \begin{bmatrix} 1 & a_{I,\Delta P} & a_{I,UI} \\ a_{\Delta P,I} & 1 & a_{\Delta P,UI} \\ a_{UI,I} & a_{UI,\Delta P} & 1 \end{bmatrix} \quad (4a)$$

$$= \begin{bmatrix} 1 & 7 & 4 \\ 1/7 & 1 & 1/3 \\ 1/4 & 3 & 1 \end{bmatrix}. \quad (4b)$$

Equation (4a) follows from Table 1a and

$$\begin{aligned} a_{j,i} &= a_{i,j}^{-1} \quad \forall i \neq j \\ a_{i,i} &= 1 \end{aligned} \quad (4c)$$

If

$$A_{O-C} X_{O-C} = X_{O-C} \Lambda_{O-C}, \quad (5)$$

represents the eigenvector/eigenvalue decomposition of reciprocal matrix A_{O-C} , the resulting priority vector ω_{O-C} becomes the Perron (or principal) eigenvector (Saaty (2003)):

$$[\omega_{O-C}]_i = \frac{[X_{O-C}]_{i, \arg\max(\Lambda_{O-C})}}{\sum_i [X_{O-C}]_{i, \arg\max(\Lambda_{O-C})}}. \quad (6)$$

Table 2a contains the reciprocal matrix A_{O-C} and the priority vector ω_{O-C} resulting from the comparative judgements (Table 1a) for the PEMFC-performance hierarchy

Table 2a: Reciprocal matrix and priority vector resulting from Table-1a hierarchy model A_{O-C} .

reference element	I	UI	ΔP	ω_i
I	1	4	7	0.705
UI	$1/4$	1	3	0.211
ΔP	$1/7$	$1/3$	1	0.084

model. The results coincide — within roundoff precision — with those reported by Bu, et al (2026) Table 12.

Table 2b contains the same information for the constituent hierarchy models $C - I$, $C - UI$, and $C - \Delta P$. The results do not coincide as closely with those from Bu, et al (2026) Table 12, which are not reported with high degrees of numerical precision. The python *scipy* library (Virtanen, et al (2020)) produced the ω_i values in Table 2b. Bu, et al (2026) do not report how their Table-12 calculations were performed. Lacking these details, exact reproduction proves difficult.

Table 2b: Reciprocal matrices and priority vectors based Table-1a hierarchy models $A_{C-\Delta P}$, A_{C-I} , A_{C-UI} .

hierarchy model	element	θ	W	H	ω_i
$C - I$	H	2	$1/3$	1	0.230
$C - I$	W	5	1	3	0.648
$C - I$	θ	1	$1/5$	$1/2$	0.1220
$C - UI$	H	$1/5$	$1/5$	1	0.0856
$C - UI$	W	3	1	5	0.618
$C - UI$	θ	1	$1/3$	5	0.230
$C - \Delta P$	H	5	7	1	0.740
$C - \Delta P$	W	$1/2$	1	$1/7$	0.0938
$C - \Delta P$	θ	1	2	$1/5$	0.1666

The mechanics illustrated by equations (4a, 5, and 6) are completely general. That ω_{O-C} is a projection onto the unit simplex Δ^3 where

$$\Delta^n = \left\{ x \in \mathbb{R}^n \mid \sum_i x_i = 1 \wedge x_i \geq 0 \forall i \right\}$$

is highly significant. All AHP calculation results are constrained to the interval $[0, 1]$. This includes both AHP-calculation results and standardization of inputs demonstrated in a subsequent subsection.

On inconsistencies in comparative judgements.

Glenn Shafer once opined, “...probability is not really about numbers; it is about the structure of reasoning...”

(quoted in Pearl (1988), p. 15). The AHP methodology contains mechanisms to identify inconsistencies and contradictions in the structure of reasoning (Saaty (2003); Pant, et al (2022)). Here, we consider the types of inconsistencies in comparative judgements from which concerns might arise. We then review consistency indices, AHP’s mechanism to quantify the degree of inconsistency. Finally, we consider alternative graph-theoretic approaches to mitigate inconsistencies in reasoning.

Types of inconsistencies comparative judgements.

Saaty (1986, 2003) expressed concern about four types of inconsistencies that might adversely affect decision making aided by the AHP. Such opportunities particularly arise in comparative judgements. Saaty (2003) seemingly suggested that accomodation of logical inconsistency in the AHP method might be more a feature than a bug: “...a modicum of inconsistency may be considered as a good thing, and forced consistency without knowledge of the precise values as an undesirable compulsion.”

The DQ framework strives for sound reasoning. Pant, et al (2022) articulates the goal of “...a decision-making process with some constraints wherein the task is to obtain the maximum benefit from the available resources to get the best achievable results.” The consequences to these goals arising from avoidable inconsistencies bear circumspectful consideration.

Circular reasoning. Expressing the first inconsistency scenario from Saaty (2003), equation (7)

$$\begin{aligned} A &\succ B, \\ B &\succ C, \text{ and} \\ C &\succ A \end{aligned} \quad (7)$$

exemplifies a preference ordering that contains circular reasoning. This obviously codifies a logical contradiction, violating the completeness and transitivity rational-preference constraints of the von-Neumann-Morgenstern axioms (Kochenderfer (2015)). Furthermore, it also violates one of de Finetti’s (1974) principles of coherence: “...no limits on the probabilities that an individual may assign, except they must not be in contradiction among themselves” (emphasis added).

American mathematician William Thurston argued that “Mathematics is not about numbers, equations, computations, or algorithms: It is about understanding” (quoted in (Cook, 2018, p. 76); emphasis added). Applying computational machinery like AHP to a contradictory logical construct does not enhance understanding. It invites confusion. Alerting producers of contradictory comparative judgements and facilitating elimination of

the circular-reasoning instances better serves the goals of sound reasoning (Spetzler, Winter, Meyer (2016)) and best-observable results (Pant, et al (2022)).

Variances between panel members' comparative judgements. MCDM methods often support group decisions. This requires eliciting comparative judgements from a range of experts. Panel members will generally provide differing ordering preferences and importance intensities. Based on Saaty (2003), Table 3 illustrates cardinal inconsistencies in ordering preferences from comparative judgements by a panel of five experts. Saaty's (2003) illustration omits importance-intensity estimates.

Aggregation of elicited probabilities from panels of experts has been explored extensively (Spetzler & Staël Von Holstein (1975); Lindley, Tversky, Brown (1979); Barons, Mascaro, Hanea (2022)). If each individual contributors' comparative judgements are free of circular reasoning — a condition satisfied in Table 3 — then established elicited-probability aggregation methods apply. Equation (4c) provides the mechanism to apply elicitation-aggregation methods when ordering preferences differ.

Some challenges may accompany the resulting probability distributions. Cooke (2026) studied the practice and observed, "Methods based on the 'wisdom of crowds' that aggregate point forecasts of multiple experts are unlikely to yield good performance in such cases, regardless of the precise method by which judgements are weighted and aggregated." The "fundamental scale" ((Saaty, 1994, Table 1)) elicits such point estimates. Li, H., et al (2025) explored elicitation of comparative judgements as interval estimates to circumvent this difficulty.

Longitudinal comparative-judgement inconsistencies. Finally, Saaty (2003) expresses concern that "...people would be required to be like robots unable to change their minds with new evidence...". Nothing elsewhere in the literature defining the AHP methodology proscribes such standard Bayesian updating (Saaty (1986, 1987, 1994)). In fact, recent research explicitly adapts AHP to Bayesian patterns of reasoning (Merrick, van Dorp, Singh (2005); Altuzarra, Moreno-Jiménez, Salvador (2007); Li, H., et al (2025)).

Transitive inconsistencies. Saaty (1986) defines a criterion for transitive consistency

$$\begin{aligned} a_{i,j} a_{j,k} &= a_{i,k} \\ \text{where } a_{i,j} &= [A_{\mathcal{H}}]_{i,j} \end{aligned} \quad (8)$$

for reciprocity matrix $A_{\mathcal{H}}$ associated with hierarchy model $\mathcal{H}$. This provides the basis for consistency indices. Now, (8) imposes a stronger constraint than the von Neumann-Morgenstern transitivity axiom previously stated in (1), which only constrains transitivity of ordering preferences.

Saaty (1986) contains an extensive set of definitions, axioms, and theorems culminating in the demonstration that, if the eigenvector/eigenvalue decomposition of a consistent reciprocal matrix $A_{\mathcal{H}}$ is

$$A_{\mathcal{H}} X_{\mathcal{H}} = X_{\mathcal{H}} \Lambda_{\mathcal{H}}, \quad (9a)$$

then

$$\lambda_{\mathcal{H}} = n \quad (9b)$$

where n denotes the order of $A_{\mathcal{H}}$ and $\lambda_{\mathcal{H}} = [\Lambda_{\mathcal{H}}]_{\text{argmax}(\Lambda_{\mathcal{H}})}$.

Transitive consistency bears important decision-theoretic consequences: "...when alternatives are scaled through paired comparisons, adding a new alternative can change the relative ranking of the old ones when the judgements are inconsistent or when several criteria are used" (Saaty (1986)).

A consistency index for reciprocal matrices.

A consequence of (9b),

$$CI = \frac{\lambda_{\mathcal{H}} - n}{n - 1} \quad (10)$$

provides a commonly used Consistency Index for a reciprocal matrix $A_{\mathcal{H}}$ (Pant, et al (2022)). From (9b), one obtains $CI = 0$ for a perfectly consistent reciprocal matrix.

A Consistency Ratio

$$CR = \frac{CI}{RI} \quad (11)$$

accompanies the consistency index. In (11), the denominator RI results from calculating CI for a large number of reciprocal matrices with randomly generated comparative judgements. Tabulated RI values are commonly repeated, such as Table 6 in Bu, et al (2026). Reproducing the common result with higher numerical precision, Donhegan & Dodd (1991) reported $RI = 0.4887$ for reciprocal-matrix order $n = 3$.

Table 4 contains the consistency analysis for the hierarchy models on which Bu, et al (2026) was based. Pant, et al (2022) cited a threshold $CR_{\mathcal{H}} \lesssim 0.1$, above which one "...questions the credibility of judgements." Comparative-judgement transitivity (8) provides the basis for credulity, here.

By this criterion, the CR_{C-UI} value in Table 4 raises questions about the consistency of reasoning in comparative judgements underlying the $C - UI$ hierarchy

Table 3: Illustration of disparate preference orderings from pairwise judgments by a panel of five experts. (After Saaty (2003))

ordering preference 1	incidence preference 1	ordering preference 2	incidence preference 2	incidence total
$A \succ B$	2	$B \succ A$	3	5
$B \succ C$	3	$C \succ B$	2	5
$A \succ C$	5	$C \succ A$	0	5

Table 4: Consistency indices for hierarchy models in PEMFC-design case study.

hierarchy model ($\mathfrak{H}$)	$\lambda_{\mathfrak{H}}$	$n_{\mathfrak{H}}$	$CI_{\mathfrak{H}}$	$CR_{\mathfrak{H}}$
$O - C$	3.032	3	0.0162	0.033
$C - I$	3.004	3	0.0018	0.004
$C - UI$	3.136	3	0.0678	0.139
$C - \Delta P$	3.014	3	0.0071	0.014

model. Not-reproduced results in Bu, et al (2026) reported a lower value for CR_{C-UI} , albeit much closer to the threshold in Pant, et al (2022).

Entropy of edge-weighted, directed acyclic graph.

Data availability.

This GitHub repository [ahp-graph-entropy](#) contains the data and the python and Neo4j logic on which this paper is based.

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